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Sequential Organozinc Formation and Negishi Cross-Coupling of Amides Catalysed by Cobalt Salt

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Abstract. Herein, a cobalt-catalysed Negishi-type cross-coupling of amide derivatives is described. Apart from being the first example of cobalt-catalysed Negishi-type coupling of amides, the process described employs a unique, simple, and cheap catalytic system to perform both the organozinc formation and the Negishi-type coupling. Indeed, the same cobalt(II) bromide salt used to form the arylzinc species from aryl bromides is then re-used to perform the cross coupling of this resulting arylzinc with *N*-benzoyl glutarimides at room temperature. The main advantages of the reaction presented are its robustness and ease of use. Indeed, the reactions of organozinc formation and Negishi-type coupling are performed without precautions toward water or oxygen.

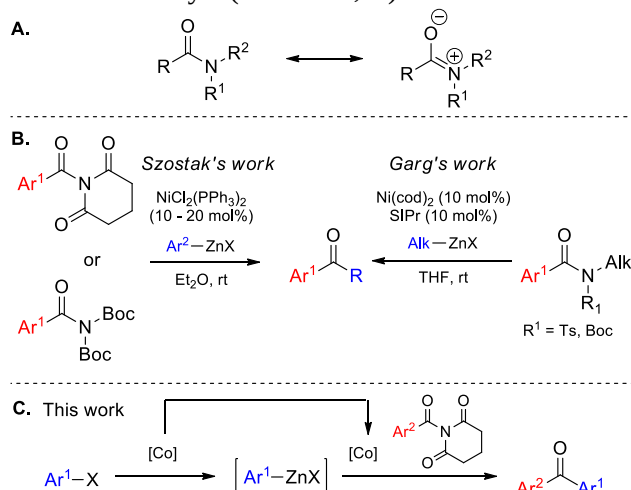
Keywords: Amides; Cobalt; Negishi-type cross-coupling; Organozinc reagent; One-pot reaction.

Amides are one of the major building blocks of life probably thanks to the high stability of the amide function.^[1] This robustness of the amidic bond, in particular toward nucleophilic attack, comes from the delocalization of the electrons of the nitrogen lone pair and those of the carbonyl moiety (Scheme 1, A).^[2] Nonetheless, the disruption of the amidic resonance through steric and/or electronic effects on the nitrogen substituents induces a distortion of the N–C amidic bond, and therefore enables its activation.^[3] Even though those so-called “twisted amides” are known from the 1940’s,^[4] it is only in 2015 that they emerged as useful electrophilic partners in transition-metal catalysed coupling reactions.^[5] Since then, both decarbonylative^[6] and non-decarbonylative cross-coupling reactions have been developed resulting in the formation of carbon–heteroatom as well as carbon–carbon bonds.^[7] Whereas Suzuki–Miyaura-type cross-coupling reactions have been quite broadly studied, only few examples of Negishi-type coupling reactions have been released. In 2016, Szostak disclosed the first nickel-catalysed Negishi-type cross-coupling reaction between highly distorted *N*-acyl glutarimides^[8] and

aryl zinc reagents.^[9] This reaction is attractive since it is performed very fast at room temperature (within 10 min for some substrates) with the bench-stable $\text{NiCl}_2(\text{PPh}_3)_2$ catalyst. Then, by tuning the catalytic system, this type of coupling has been broadened to *N,N*-di-Boc^[10] and *N*-acyl succinimide.^[11] Meanwhile, Garg published the first Negishi-type cross-coupling reaction allowing the alkylation of the less distorted *N*-alkyl, Ts or *N*-alkyl, Boc amide derivatives^[12] catalysed by $\text{Ni}(\text{cod})_2/\text{SIPr}$ (Scheme 1, B).^[13]

Although organozinc compounds present high nucleophilicity and offer an excellent functional group tolerance, their preparation can be delicate to handle. Indeed, they are mostly prepared either in one step by insertion of activated zinc powder in the presence of LiCl ^[14] or in two steps by transmetalation between a beforehand prepared organolithium or organomagnesium reagent and zinc salt. In both cases, their formation requires highly anhydrous and anaerobic conditions. In 2003, we revealed an easy to handle cobalt-catalysed formation of aryl zinc compounds, necessitating no particular precautions,^[15] and their reactivity in Negishi-type reactions^[16–17] with acid chloride or anhydride derivatives,^[18] among others.^[19] This method is quite efficient requiring no additional ligand for the reaction to perform at room temperature. Moreover, the process is easy to use and robust since the reaction can be run in solvents neither particularly dried nor degassed and under air, without decreasing the yield of the transformation. Recently, we first revealed the ability of cobalt salt to activate the amidic bond of twisted amides in order to achieve the non-thermodynamically favored amide to ester transformation.^[20] Taking all this into account, we therefore have been curious to develop the first cobalt-catalysed Negishi-type cross-coupling reaction involving amides as coupling partners and offering a simple process and neither anhydrous nor anaerobic conditions. Furthermore, it could be of great interest to achieve a sequential method in which the unique cobalt salt is used as catalyst, not only for the preparation of the zinc species, but also to perform

the Negishi-type cross-coupling reaction, without any additional catalyst (Scheme 1, C).



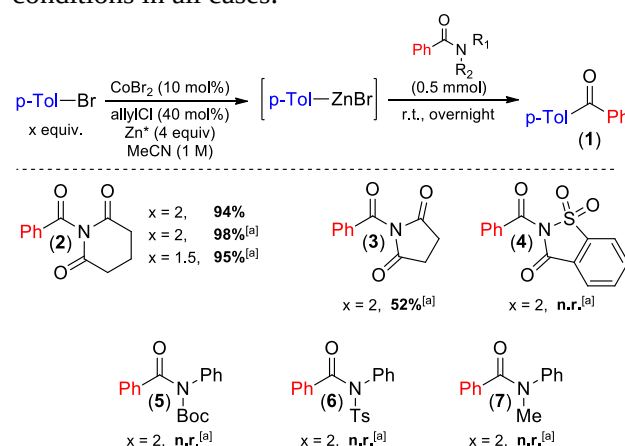
Scheme 1. **A.** Amide resonance; **B.** Previous Negishi-type cross-coupling of amides; **C.** This work.

We investigated this sequential strategy involving Negishi-type coupling on *N*-benzoyl glutarimide (**2**). This distorted amide has already shown high reactivity in cross-coupling reaction and especially in Negishi-type reaction.^[8,9] We decided to form the organometallic coupling partner, *p*-tolylzinc derivative according to our procedure, using simple cobalt (II) bromide salt as catalyst in presence of an excess of metallic zinc powder activated by 40 mol% of allyl chloride and traces of TFA in acetonitrile (1 M).^[15] The formation of the *p*-tolylzinc species was monitored by GC, using tetradecane as internal standard, following the amount of aromatic iodide formed after iodination of a sample of the mixture. After approximately 1 h of reaction, the *p*-bromotoluene was fully converted into, for the most part, zinc compound (on average 70-80% of *p*-iodotoluene), the rest being a mix of dimers and toluene. Subsequently, 1 mL of this mixture (2 equivalents of aryl bromide corresponding to 1.4 equivalents of zinc compound) was added into a vial containing *N*-benzoyl glutarimide (**2**; 0.5 mmol) and the resulting mixture was stirred overnight at room temperature. We were pleased to observe a near total conversion of the amide starting material (**2**) into the corresponding diaryl ketone (**1**) in 94% yield (Scheme 2). We also tried to filter the aforementioned organozinc compound solution in order to get rid of the zinc powder in excess before performing the coupling reaction. The resulting orange solution containing arylzinc species and homogenous cobalt catalyst was then added again onto the *N*-benzoyl glutarimide (**2**). This also led to the corresponding diaryl ketone (**1**) in near-quantitative yield (Scheme 2). It is important to note that with both procedures, the coupling reaction is really straightforward, leading only to the desired diaryl ketone product **1**. However, the second process, involving the simple filtration step of the organozinc species, mostly

simplifies the treatment of the reaction. Indeed, in this case, no workup is needed and the whole reaction mixture can be subjected to column chromatography to give the product in quantitative yield.

According to previous literature by Szostak and Garg,^[9-13] imides, including *N*-benzoyl glutarimides, do not react with organozinc reagents. However, these reactions were carried out in THF, toluene, or ether. We therefore decided to verify whether this is valid for acetonitrile, the solvent used to perform our coupling reaction. To this end, we first prepared the *p*-tolylzinc derivative by transmetalation of zinc chloride salt with both the corresponding Grignard and lithium reagents. Then, for both organozinc species, 2 equivalents of the arylzinc solution were added into a vial containing a solution of the *N*-benzoyl glutarimide (**2**) in acetonitrile (1 M) on one hand, and a vial containing a solution of the *N*-benzoyl glutarimide (**2**) and CoBr₂ (10 mol%) in acetonitrile (1 M) on the other hand. Regardless of which organozinc species was used, the reactions performed without cobalt salt showed no diaryl ketone product (**1**) whereas the ones carried out with 10 mol% of CoBr₂ led to more than 90% yield of the desired product (**1**), hence validating the necessity of the cobalt salt to achieve this Negishi-type coupling.

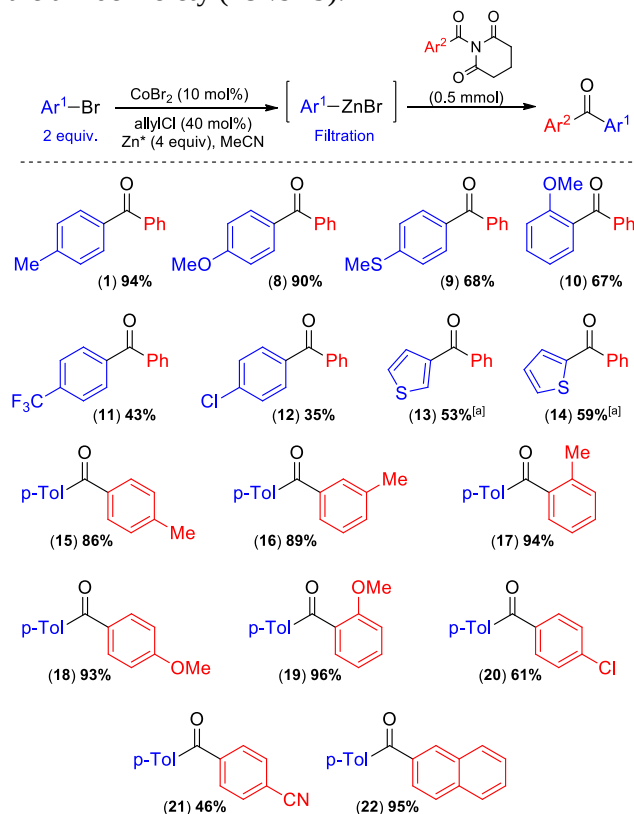
We then tried to lower the amount of the arylzinc coupling partner up to the stoichiometry (1.5 equivalents of aryl bromide corresponding to 1.05 equivalents of zinc compound). To our best delight, no significant decrease of the yield was observed (Scheme 2) confirming the high efficiency of the coupling even in stoichiometric amount of the organozinc reagent. However, based on our previous publications on arylzinc formation,^[15] all zinc compounds are not prepared with equal efficiency. For simplicity and considering the low cost of the cobalt catalyst used, we decided to investigate the scope of the reaction by using 2 equivalents of aryl bromides compared to the amide to ensure optimal conditions in all cases.



Scheme 2. Study of the sequential organozinc formation and Negishi cross-coupling of amides. The yields were determined by GC analysis using dodecane as internal standard. ^[a] The solution of organozinc was filtered prior to use.

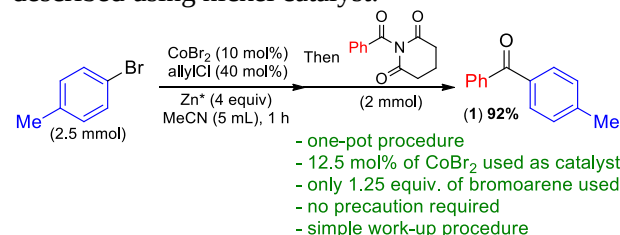
Finally, we evaluated the range of distorted amides which can be activated by this process. Only highly distorted *N*-acyl glutarimide (**2**) and succinimide (**3**) can actually be converted using this catalytic process. Less-distorted amides (**4-7**) such as *N*-Boc (**5**) or *N*-Ts (**6**) are completely recovered and therefore unreactive in the reaction conditions. This selectivity is synthetically interesting since it thus allows the coupling of *N*-acyl glutarimide without modification of the other aforementioned twisted amides (Scheme 2).

With these conditions in hand, we then investigated the scope of the reaction (Scheme 3). First, different arylzinc were formed and engaged in the Negishi-type cross-coupling with *N*-benzoyl glutarimide (**2**). Arylzinc carrying an electron-donating group such as methyl, methoxy or even a catalyst poisoning thiomethyl group show good reactivity and lead to the corresponding diaryl ketone (**1** and **8-10**) in good to excellent yields. For the organozinc bearing an electron-withdrawing group as well as heteroarylzinc species, the diaryl ketones (**11-14**) were obtained in moderate yields. In terms of amide diversity, the same trend was observed. Indeed, *N*-benzoyl glutarimide with an electron-withdrawing group (**20-21**) lead to moderate yields while those with an electron-donating group (**15-19**) lead to excellent yields. Finally, steric hindrance seems to have an influence on the arylzinc (**10** vs **8**) but not on the amide moiety (**18** vs **19**).



Scheme 3. Scope of the reaction. Isolated yields were noted. ^[a] 20 mol% of CoBr₂ and 80 mol% of allyl chloride were used.

In order to provide a really useful and efficient process, a one-pot procedure performing both reactions was attempted. This one-pot procedure allowed to decrease the amount of bromoarene used to 1.25 equivalents as well as the catalyst loading to 12.5 mol%. After simple filtration of the reaction mixture through a pad of celite and purification on column chromatography, the corresponding diaryl ketone was obtained in pure form with an excellent yield of 92% (Scheme 4). The development of this procedure offers a cheap and easy to handle alternative to the reductive cross-coupling previously described using nickel catalyst.^[21]



Scheme 4. One-pot procedure involving organozinc formation and Negishi cross-coupling

In conclusion, we have developed the first cobalt-catalysed Negishi-type cross-coupling reaction involving amides as coupling partners. The process uses a simple and cheap catalytic system. First, the cobalt (II) bromide salt catalyses both the preparation of the zinc species as well as the Negishi-type cross-coupling reaction. Second, the reaction is quite efficient since no additional ligand for the cobalt catalyst is needed to carry out the sequential organozinc formation and the coupling reaction at room temperature. In addition, the set-up of the reaction is robust and easy to handle since it necessitates neither distilled solvent nor aqueous work-up, leading to the pure cross-coupled product after simple column chromatography. Finally, this method shows its great usefulness by reaching the pure product in one-pot procedure with lowered amounts of zinc species and catalyst.

Experimental Section

General procedure for the organozinc formation and Negishi-type cross-coupling of amides

A 20 mL vial was charged with CoBr₂ (109 mg, 0.5 mmol, 0.1 equiv.) and zinc powder (981 mg, 20 mmol, 4 equiv.) and sealed. MeCN (5 mL) was then added, followed by allyl chloride (163 μ L, 2 mmol, 0.4 equiv.) and TFA (50 μ L). After 15 min at room temperature, the bromoarene (5 mmol) was added in one portion and the reaction was stirred at room temperature until complete conversion of the bromoarene (typically 1 hour). 1 mL of the resulting slurry was taken *via* a syringe equipped with a filter. The clear orange solution was then added into a 20 mL vial containing the amide derivative (0.5 mmol) and the resulting solution was stirred overnight at room temperature. A precipitate appeared during the reaction. The resulting mixture was taken and injected directly on a silica gel column chromatography for purification (eluent: petroleum ether / ethyl acetate).

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References

- [1] a) A. Greenberg, C. M. Breneman, J. F. Liebman in *The Amide Linkage: Structural Aspects in Chemistry, Biochemistry, and Materials Science*. New York: Wiley-Interscience; **2000**; b) V. R. Pattabiraman, J. W. Bode, *Nature* **2011**, 480, 471; c) A. A. Kaspar, J. M. Reichert, *Drug Discovery Today* **2013**, 18, 807.
- [2] C. R. Kemnitz, M. J. Loewen, *J. Am. Chem. Soc.* **2007**, 129, 2521-2528.
- [3] a) C. Liu, M. Szostak, *Chem. Eur. J.* **2017**, 23, 7157-7173; b) D. Kaiser, A. Bauer, M. Lemmerer, N. Maulide, *Chem. Soc. Rev.* **2018**, 47, 7899-7925.
- [4] M. Szostak, J. Aubé, *Chem. Rev.* **2013**, 113, 5701-5765.
- [5] a) X. Li, G. Zou, *Chem. Commun.* **2015**, 51, 5089-5092; b) G. Meng, M. Szostak, *Org. Lett.* **2015**, 17, 4364-4367; c) G. Meng, M. Szostak, *Angew. Chem. Int. Ed.* **2015**, 54, 14518-14522; d) L. Hie, N. F. Fine Nathel, T. K. Shah, E. L. Baker, X. Hong, Y.-F. Yang, P. Liu, K. N. Houk, N. K. Garg, *Nature* **2015**, 524, 79.
- [6] For review see: C. Liu, M. Szostak, *Org. Biomol. Chem.* **2018**, 16, 7998-8010.
- [7] For general reviews see: a) J. E. Dander, N. K. Garg, *ACS Catal.* **2017**, 7, 1413-1423; b) G. Meng, S. Shi, M. Szostak, *Synlett* **2016**, 27, 2530-2540; c) S. Adachi, N. Kumagai, M. Shibasaki, *Tetrahedron Lett.* **2018**, 59, 1147-1158; d) R. Takise, K. Muto, J. Yamaguchi, *Chem. Soc. Rev.* **2017**, 46, 5864-5888.
- [8] For review see: G. Meng, M. Szostak, *Eur. J. Org. Chem.* **2018**, 2018, 2352-2365.
- [9] S. Shi, M. Szostak, *Chem. Eur. J.* **2016**, 22, 10420-10424.
- [10] S. Shi, M. Szostak, *Org. Lett.* **2016**, 18, 5872-5875.
- [11] S. Shi, M. Szostak, *Synthesis* **2017**, 49, 3602-3608.
- [12] Y. Bourne-Branchu, C. Gosmini, G. Danoun, *Chem. Eur. J.*, in press, DOI: 10.1002/chem.201802635.
- [13] B. J. Simmons, N. A. Weires, J. E. Dander, N. K. Garg, *ACS Catal.* **2016**, 6, 3176-3179.
- [14] A. Krasovskiy, V. Malakhov, A. Gavryushin, P. Knochel, *Angew. Chem. Int. Ed.* **2006**, 45, 6040-6044.
- [15] a) H. Fillon, C. Gosmini, J. Périchon, *J. Am. Chem. Soc.* **2003**, 125, 3867-3870; b) I. Kazmierski, C. Gosmini, J.-M. Paris, J. Périchon, *Tetrahedron Lett.* **2003**, 44, 6417-6420.
- [16] For reviews in cobalt-catalysed cross-couplings see: a) G. Cahiez, A. Moyeux, *Chem. Rev.* **2010**, 110, 1435-1462; b) C. Gosmini, J.-M. Bégouin, A. Moncomble, *Chem. Commun.* **2008**, 3221-3233; c) C. Gosmini, A. Moncomble, *Isr. J. Chem.* **2010**, 50, 568-576.
- [17] For selected reviews on Negishi cross-couplings see: a) D. Haas, J. M. Hammann, R. Greiner, P. Knochel, *ACS Catal.* **2016**, 6, 1540-1552; b) V. B. Phapale, D. J. Cárdenas, *Chem. Soc. Rev.* **2009**, 38, 1598-1607; c) E. Negishi, *Acc. Chem. Res.* **1982**, 15, 340-348.
- [18] a) I. Kazmierski, M. Bastienne, C. Gosmini, J.-M. Paris, J. Périchon, *J. Org. Chem.* **2004**, 69, 936-942; b) H. Fillon, C. Gosmini, J. Périchon, *Tetrahedron* **2003**, 59, 8199-8202.
- [19] For selected publications see: a) M. Corpet, X.-Z. Bai, C. Gosmini, *Adv. Synth. Catal.* **2014**, 356, 2937-2942; b) J.-M. Bégouin, M. Rivard, C. Gosmini, *Chem. Commun.* **2010**, 46, 5972-5974; c) M. Amatore, C. Gosmini, *Chem. Commun.* **2008**, 5019-5021.
- [20] Y. Bourne-Branchu, C. Gosmini, G. Danoun, *Chem. Eur. J.* **2017**, 23, 10043-10047.
- [21] S. Ni, W. Zhang, H. Mei, J. Han, Y. Pan, *Org. Lett.* **2017**, 19, 2536-2539.

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